Exact solution of the arbitrary-noise optimal-signals problem

Candidate solution packet for arXiv:2002.03974

13 August 2026

Status. Candidate full solution to Problem 1 of Ambrus–Bai–Hou [1], pending expert review. The source solves the zero-noise case and the positive-noise problem restricted to uniform systems. The theorem below gives the exact value and all maximizers for every noise level.

1 Source question

The source is G. Ambrus, B. Bai, and J. Hou, *Uniform tight frames as optimal signals* [1]. It is arXiv:2002.03974v3, Problem 1 on PDF page 2 and the complete statement and the authors' scope note are shown below.

Problem 1. *Assume that $d \geqslant 2$, $N \geqslant d$, $0 < c_1 < c_2$ are positive bounds, and $\sigma \geqslant 0$ is a constant. Determine the quantity*

$$\max_{\substack{v_1,\ldots,v_N \in \mathbb{R}^d \\ c_1 \leqslant |v_i|^2 \leqslant c_2 \ \forall i}} \min_{1 \leqslant k \leqslant N} \log \left(1 + \frac{|v_k|^2}{\sigma^2 + \sum_{l \neq k} \langle v_k, v_l \rangle^2} \right). \tag{2}$$

In the present article, we solve Problem 1 in the special case $\sigma = 0$, and obtain a stability version for small values of $\sigma > 0$. The latter is essential for practical applications in signal processing.

Thus, for integers $d \geq 2$ and $N \geq d$, bounds $0 < c_1 < c_2$, and $\sigma \geq 0$, one must determine

$$\max_{\substack{v_1,\ldots,v_N \in \mathbb{R}^d \\ c_1 \leq \|v_i\|^2 \leq c_2}} \min_{1 \leq k \leq N} \log \left(1 + \frac{\|v_k\|^2}{\sigma^2 + \sum_{\ell \neq k} \langle v_k, v_\ell \rangle^2} \right). \tag{1}$$

The source uses the convention $1/0 = +\infty$.

2 Definitions and exact answer

A family $u_1, \ldots, u_N \in \mathbb{R}^d$ is a *unit-norm tight frame* (UNTF) if

$$\|u_i\| = 1 \quad (1 \le i \le N), \qquad \sum_{i=1}^{N} u_i \otimes u_i = \frac{N}{d} I_d.$$

Real UNTFs exist for every $N \ge d$; see [2, 1].

Theorem 1 (Complete arbitrary-noise formula). *Assume $N > d$ and put*

$$\alpha = \frac{N-d}{d}, \qquad c_\sigma = \min\left\{c_2, \max\left\{c_1, \frac{\sigma}{\sqrt{\alpha}}\right\}\right\}, \qquad \tau_\sigma = \frac{\sigma^2}{c_\sigma} + \alpha c_\sigma.$$

Then the value of (1) *is*

$$\boxed{\log\left(1 + \frac{1}{\tau_\sigma}\right)}. \tag{2}$$

Moreover, a family $(v_i)_{i=1}^N$ is a maximizer if and only if

$$v_i = \sqrt{c_\sigma}\, u_i \quad (1 \le i \le N) \tag{3}$$

for a UNTF $(u_i)_{i=1}^N$.

Equivalently,

$$\tau_\sigma = \begin{cases} \sigma^2/c_1 + \alpha c_1, & 0 \le \sigma \le c_1\sqrt{\alpha}, \\ 2\sigma\sqrt{\alpha}, & c_1\sqrt{\alpha} \le \sigma \le c_2\sqrt{\alpha}, \\ \sigma^2/c_2 + \alpha c_2, & \sigma \ge c_2\sqrt{\alpha}. \end{cases}$$

At the common endpoints the adjacent expressions agree.

For completeness, when $N = d$ and $\sigma > 0$, the value is $\log(1 + c_2/\sigma^2)$ and the maximizers are precisely the orthogonal bases whose vectors all have squared norm c_2. When $N = d$ and $\sigma = 0$, the value is $+\infty$ and the maximizers are precisely the orthogonal bases with arbitrary squared norms in $[c_1, c_2]$.

3 Proof intuition

Taking reciprocals turns the max–min signal-to-noise problem into a minimax problem. The obstruction in the nonuniform case is that user i sees the other squared norms with asymmetric coefficients. The useful choice is to average the reciprocal SINR with weight $1/\|v_i\|^2$. For an unordered pair $\{i, j\}$, the two directed contributions then combine into

$$\frac{\|v_j\|^2}{\|v_i\|^2} + \frac{\|v_i\|^2}{\|v_j\|^2} \ge 2.$$

This erases the norm asymmetry. The surviving angular sum is exactly the frame potential controlled by the Welch bound. Cauchy–Schwarz then replaces all inverse squared norms by their mean, reducing the entire optimization to the one-variable function $\sigma^2/c + \alpha c$. A UNTF at its clipped minimizing value of c makes every inequality an equality.

4 Proof

We give the central estimate separately.

Lemma 1 (Inverse-norm weighted Welch bound)**.** *Let $N \geq d$, let $a_i \in [c_1, c_2]$, and let $u_i \in S^{d-1}$. Set*

$$D_i = \frac{\sigma^2}{a_i} + \sum_{j \neq i} a_j \langle u_i, u_j \rangle^2, \qquad x_i = \frac{1}{a_i}, \qquad \bar{x} = \frac{1}{N} \sum_{i=1}^{N} x_i.$$

With $\alpha = (N-d)/d$, one has

$$\max_i D_i \geq \frac{\sum_i x_i D_i}{\sum_i x_i} \geq \frac{\sigma^2 \sum_i x_i^2 + N\alpha}{\sum_i x_i} \geq \sigma^2 \bar{x} + \frac{\alpha}{\bar{x}}. \tag{4}$$

Proof. The first inequality is a positive weighted average. Expanding its numerator and grouping the two directed terms belonging to each unordered pair gives

$$\begin{aligned} \sum_i x_i D_i &= \sigma^2 \sum_i x_i^2 + \sum_{i<j} \left(\frac{a_j}{a_i} + \frac{a_i}{a_j} \right) \langle u_i, u_j \rangle^2 \\ &\geq \sigma^2 \sum_i x_i^2 + 2 \sum_{i<j} \langle u_i, u_j \rangle^2. \end{aligned}$$

For $T = \sum_i u_i \otimes u_i$, one has $\operatorname{tr} T = N$ and hence

$$2 \sum_{i<j} \langle u_i, u_j \rangle^2 = \|T\|_{\mathrm{HS}}^2 - N \geq \frac{(\operatorname{tr} T)^2}{d} - N = \frac{N(N-d)}{d} = N\alpha.$$

This is the usual Welch bound, here proved directly from Cauchy–Schwarz for the eigenvalues of T. Finally, $\sum_i x_i^2 \geq N\bar{x}^2$ and $\sum_i x_i = N\bar{x}$, proving the last step of (4). $\square$

Proof of Theorem 1. Write $v_i = \sqrt{a_i}\, u_i$. The reciprocal of user i's signal-to-noise ratio is exactly

$$\frac{\sigma^2 + \sum_{j \neq i} \langle v_i, v_j \rangle^2}{\|v_i\|^2} = D_i.$$

Consequently, maximizing the minimum signal-to-noise ratio is equivalent to minimizing $\max_i D_i$. Since $x_i \in [1/c_2, 1/c_1]$, also $\bar{x} \in [1/c_2, 1/c_1]$. Lemma 1 gives

$$\begin{aligned} \max_i D_i &\geq \min_{x \in [1/c_2, 1/c_1]} \left(\sigma^2 x + \frac{\alpha}{x} \right) \\ &= \min_{c \in [c_1, c_2]} \left(\frac{\sigma^2}{c} + \alpha c \right) = \frac{\sigma^2}{c_\sigma} + \alpha c_\sigma = \tau_\sigma. \end{aligned} \tag{5}$$

The last scalar function decreases up to $c = \sigma/\sqrt{\alpha}$ and increases thereafter, which yields the stated clipping and piecewise formulas.

Let (u_i) be a UNTF and set $v_i = \sqrt{c_\sigma} u_i$. Its frame operator is $(Nc_\sigma/d)I_d$, so for every i,

$$\sum_{j \neq i} \langle v_i, v_j \rangle^2 = \langle v_i, \left(\sum_j v_j \otimes v_j \right) v_i \rangle - \|v_i\|^4 = \frac{N-d}{d} c_\sigma^2 = \alpha c_\sigma^2.$$

Thus every D_i equals τ_σ, proving sharpness of (5) and the value (2).

It remains to characterize equality. Suppose first that $\sigma > 0$. Equality in the last step of Lemma 1 forces equality in $\sum_i x_i^2 \geq N\bar{x}^2$, hence all x_i and all a_i are equal. The strict convexity of $\sigma^2/c + \alpha c$ forces their common value to be c_σ. Equality in the Welch step forces $T = (N/d)I_d$, so the normalized vectors form a UNTF.

If $\sigma = 0$ (still with $N > d$), the last expression in (4) is $\alpha/\bar{x} \geq \alpha c_1$. Equality forces $\bar{x} = 1/c_1$; because every $x_i \leq 1/c_1$, this means $a_i = c_1 = c_\sigma$ for every i. Equality in the Welch step again gives a UNTF. Conversely, the displayed construction proves that every family in (3) is a maximizer. □

The case $N = d$. Here $\alpha = 0$. For $\sigma > 0$, the same chain gives

$$\max_i D_i \geq \frac{\sigma^2 \sum_i x_i^2}{\sum_i x_i} \geq \sigma^2 \bar{x} \geq \frac{\sigma^2}{c_2}.$$

Equality forces $a_i = c_2$ for every i, and equality in the nonnegative pair contribution forces the u_i to be pairwise orthogonal. Such a scaled orthogonal basis attains the bound. If $\sigma = 0$, $\max_i D_i$ is zero exactly when all pairwise inner products vanish, and the lengths are unrestricted within the given interval. This proves the boundary claims.

5 Verification

The companion script `code/verify_bound.py` performs two checks. With a fixed seed it tested all four inequalities in (4) and (5) on 120,000 random configurations in six dimension/size pairs, with random norm bounds and noise levels (including zero noise). It also constructed explicit real harmonic UNTFs for eight dimension/size pairs and verified equality in 48 parameter regimes spanning both clipping thresholds. The command was

```
conda run --no-capture-output -n sandbox python code/verify_bound.py
```

and all checks passed. Small $d = 2$, $N = 3, 4, 5$ differential-evolution searches over lengths and angles also returned the theorem's value across small, intermediate, and large noise. These computations test for algebraic or edge-case mistakes; they are not used in the proof.

6 Novelty check, scope, and reviewer focus

The bounded check was performed on 13 August 2026. The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2002.03974, the exact title, and the core max–min/tight-frame terms. Web and arXiv searches used the exact title and id, the phrase "extremal vector systems are not necessarily uniform," and formula-level combinations of "max min," "noise," and "tight frame." OpenAlex's citation record for the published DOI 10.1016/j.aam.2021.102219 listed one citing work, *Robustness of Frames and Totally Nonsingular Matrices* (2026), whose stated topic is unrelated to this extremum problem. No later paper claiming the arbitrary-noise solution was found. This is a bounded search, not a proof of novelty.

The result answers the finite-dimensional real problem exactly as stated. It does not address complex frames, altered interference exponents, unequal user-specific noise, or additional coding constraints. A reviewer should check especially (i) the inverse-norm weighting identity, (ii) that every inequality in the sharp chain must be an equality at a maximizer, and (iii) the separate equality arguments for $\sigma = 0$ and $N = d$. The source states informally that positive-noise extremizers need not be uniform [1, p. 4]; Theorem 1 implies the contrary for Problem 1 as printed, so this point merits direct expert scrutiny.

References

[1] G. Ambrus, B. Bai, and J. Hou, *Uniform tight frames as optimal signals*, Advances in Applied Mathematics **129** (2021), 102219; arXiv:2002.03974v3.

[2] J. J. Benedetto and M. Fickus, *Finite normalized tight frames*, Advances in Computational Mathematics **18** (2003), 357–385.